Evaluating and Training Large Language Models: What Can (and Cannot) Item Parameters Tell Us?

Stage-1 Open-Ended Psychometric Discovery & Discussion · October 26 Research Group Seminar

The Superficial ML / LLM Consensus (CTT)

Standard LLM evaluation and post-training pipelines rely almost exclusively on **Classical Test Theory (CTT) aggregate sum-scores**:

- **In Evaluation**: Report mean single-draw accuracy ($\mathrm{Pass@1}$) or compress a 100-model leaderboard into a unidimensional ability scalar $\hat{\theta}$ (*tinyBenchmarks*, *metabench*).
- **In RL Post-Training**: Plug static population IRT parameters $(a_j^{\mathrm{pop}}, b_j^{\mathrm{pop}})$ into outcome RL (`GRPO`) hoping for an automatic accuracy boost—and conclude IRT "doesn't work" when $\mathrm{Pass@1}$ saturates.

Our Psychometric Thesis (Why IRT Matters Here)

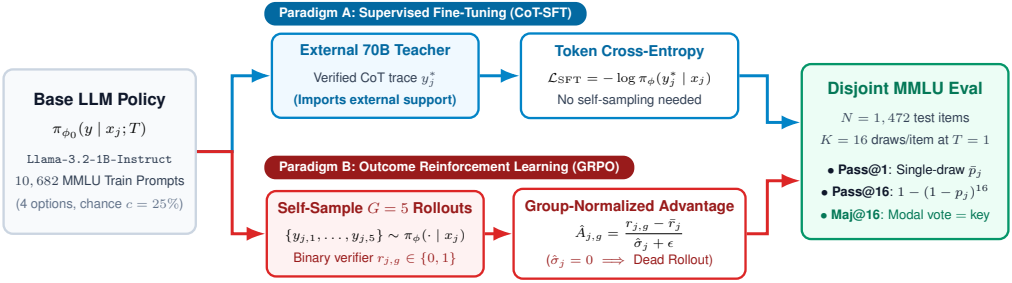
Even when static population (a_j, b_j) yields **zero headline** $\mathrm{Pass@1}$ **gain** in RL training, psychometric measurement theory exposes what aggregate ML averages are blind to:

1. **Construct Validity & Domain DIF**: Why 1D adaptive testing hits a 2.4% floor ($5\times$ loading bias toward factual recall over math).
2. **Between- vs. Within-Respondent Frame**: Why leaderboard (a_j, b_j) is orthogonal ($r = 0.072$) to single-model learning trajectories.
3. **PRF Slope Rotation & 3PL Distractors**: Why `GRPO` $\mathrm{Pass@1}$ gains are temperature sharpening ($\tau > 1$) around the $c = 25\%$ guessing floor, locking in wrong modal distractors.
4. **Local Independence & Format DIF**: Cross-seed trajectory rigidity ($r = 0.81$) and $4\times$ option-position bias ($D = 10.67$).

Framing for Today's Discussion: This is **not** a closed-end engineering report claiming a leaderboard win. It is a **Stage-1 diagnostic exploration** showing how familiar psychometric concepts (*2PL/3PL, Construct Validity, Person Response Functions, Local Independence, DIF, Fisher Information*) uncover structural mechanisms in LLMs that standard ML metrics miss.

Setup I: How LLMs Are Post-Trained and Evaluated on Multiple-Choice Benchmarks

Establishing Shared Notation Across Machine Learning and Psychometrics



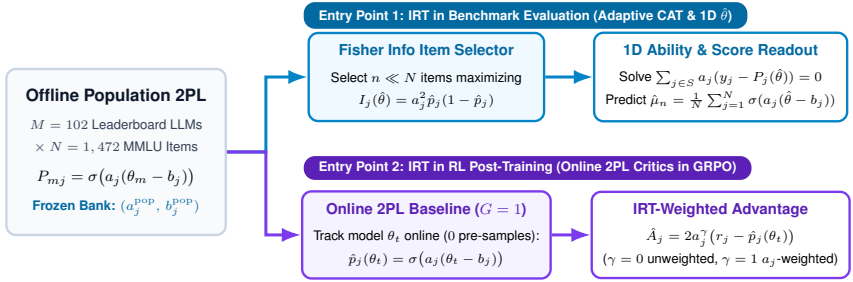
1. CoT-SFT (External Teacher)
Minimizes token cross-entropy on verified reasoning traces y_j^* from a 70B teacher (Rafailov et al., 2023).
• **Imports external support**: updates policy even when Base model has $p_{\mathrm{base},j} = 0$.

2. GRPO (Self-Sampled RL)
Samples $G = 5$ rollouts from current policy π_ϕ, rewards binary outcome $r_{j,g} \in \{0, 1\}$, normalizes within-group (Shao et al., 2024).
• **Zero-variance trap**: if all G draws fail or succeed ($\hat{\sigma}_j = 0$), gradient vanishes ($\hat{A}_{j,g} = 0$).

3. Three Disjoint Readouts
At test time ($K = 16$ i.i.d. draws/item; Chen et al., 2021):
• $\mathbf{Pass@1} = \hat{p}_j$: single-draw accuracy.
• $\mathbf{Pass@16} = 1 - (1 - \hat{p}_j)^{16}$: multi-draw coverage.
• $\mathbf{Maj@16} = \mathbb{I}[\arg\max_o \hat{p}_j(o) = o_j^*]$: does the **modal answer** match the key?

Setup II: Where Does Offline Population 2PL IRT Enter?

Two Entry Points: Adaptive Benchmark Evaluation vs. RL Advantage Critics



Entry Point 1: Efficient Benchmark Evaluation (Literature)

Fit a 2PL item bank $P_{mj} = \sigma(a_j(\theta_m - b_j))$ across $M = 102$ Open-LLM Leaderboard models $\times$ $N = 1,472$ MMLU test items (Maia Polo et al., ICML 2024; Kipnis et al., ICLR 2025).

- **Promise**: Select $n \ll N$ high-Fisher-information items $I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$ via Computerized Adaptive Testing (`CAT`) to recover full-benchmark score from 32–128 items.

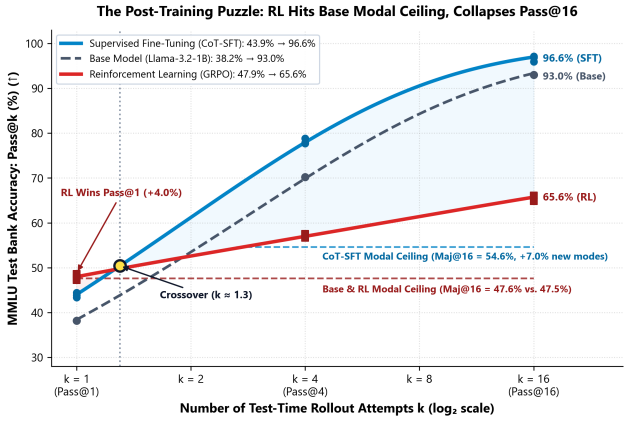
Entry Point 2: Replacing Rollouts with Online 2PL Critics

Standard `GRPO` spends $G = 5$ rollouts/item just to estimate baseline difficulty $\bar{r}_j$ (Hofmann et al., COLM 2025). Can frozen population $(a_j^{\text{pop}}, b_j^{\text{pop}})$ provide a **zero-rollout ($G = 1$) baseline** and focus updates on high-a_j items?

$$\hat{A}_j = 2\, a_j^{\gamma}\left(r_j - \sigma(a_j(\theta_t - b_j))\right), \qquad \gamma \in \{0, 1\}$$

Module 0: The Empirical Puzzle — Single-Draw Gain vs. Multi-Draw Collapse

Why Aggregate Pass@1 Hides a Fundamental Support-vs.-Sharpening Divergence



What Standard ML Sees (Pass@1 Only):

Both post-training paradigms look like clear wins over `Base` (38.2%), with `GRPO` beating `CoT-SFT` by $+4.0$ pp at $k = 1$:

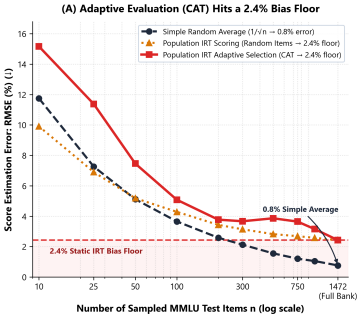
- `Base (Llama-3.2-1B)`: $\mathbf{38.2}\%$
- `CoT-SFT`: $\mathbf{43.9}\%$ ($+5.7$ pp)
- `GRPO (RL)`: $\mathbf{47.9}\%$ ($+9.7$ pp)

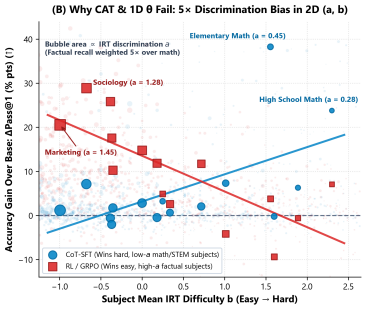
What Multi-Draw & Modal Readouts Reveal:

- **Immediate Crossover ($k \approx 1.3$)**: By $k = 16$ draws, `GRPO` collapses to $\mathbf{65.6}\%$—$\mathbf{27.4}$ pp **below** `Base` (93.0%) and $\mathbf{31.0}$ pp **below** `CoT-SFT` (96.6%).
- **Modal-Key Ceiling ($\mathrm{Maj@16}$)**: Majority-vote accuracy is **identical** between `Base` ($\mathbf{47.6}\%$) and `GRPO` ($\mathbf{47.5}\%$), whereas `CoT-SFT` reaches $\mathbf{54.6}\%$ ($+7.0\%$ new modes).
- `GRPO`'s single-draw $\mathrm{Pass@1}$ (47.9%) simply sharpens probability onto the Base model's pre-existing modal keys (**0% net new modes learned**).

Module 1 (Evaluation): Construct Validity & Domain Bias in Population 2PL

Why Adaptive CAT Hits a 2.4% Bias Floor When Ranking Post-Trained Models





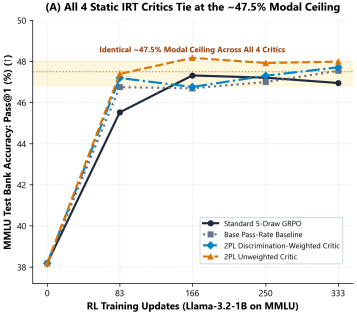
1. Adaptive CAT Bias Floor (2.4%)
Simple random averaging decays as $1/\sqrt{n} \rightarrow \mathbf{0.8}\%$ RMSE (Panel A), whereas population 2PL scoring and CAT plateau at a $\mathbf{2.4}\%$ error floor even at $n = 1,472$.

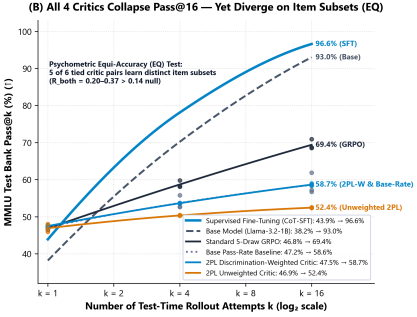
2. Construct Validity: $5\times$ Loading Skew
In the 102-LLM bank, factual recall subjects (*Marketing* $a = 1.45$, *Sociology* $a = 1.28$) have $5\times$ **higher** a_j than math (*HS Math* $a = 0.28$). Since $I_j(\theta) \propto a_j^2$, CAT oversamples memory $25\times$!

3. Domain DIF Under Post-Training
GRPO boosts easy, high-a factual domains, while CoT-SFT lifts hard, low-a math domains (Panel B)—violating 1D measurement invariance.

Module 2A (RL Training): Do Population 2PL Critics Help GRPO?

Aggregate Pass@1 Equivalence Masks High-k Collapse and 5/6 Equi-Accuracy Divergence





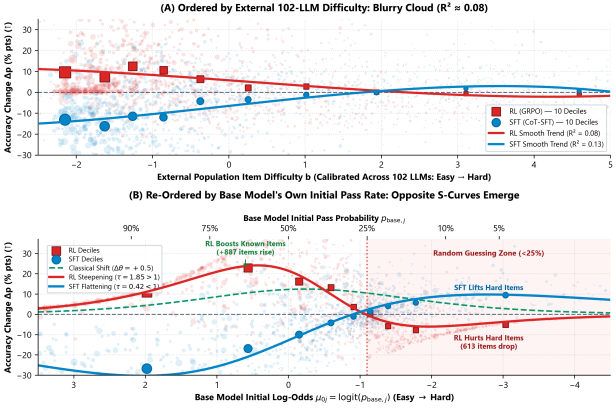
1. Identical $\sim 47.5\%$ $\mathrm{Pass@1}$ Ceiling
Across 333 RL updates, all 4 critics ($G = 5$ GRPO, Base Pass-Rate, 2PL Unweighted, 2PL a_j-Weighted) saturate inside the exact same $\mathbf{46.8\%}$–$\mathbf{47.5\%}$ modal band (Panel A).

2. Severe Multi-Draw $\mathrm{Pass@16}$ Collapse
Single-rollout critics ($G = 1$) save $5\times$ sampling cost, but weighting by a_j^{pop} worsens multi-draw collapse at $k = 16$ ($\mathbf{58.7\%}$ and $\mathbf{52.4\%}$ vs. 93.0% Base, Panel B).

3. Equi-Accuracy (EQ) Divergence
Conditioning on tied $\mathrm{Pass@1}$, **5 of 6 critic pairs** solve significantly different item subsets ($R_{\mathrm{both}} = 0.20$–$0.37 > 0.14$ null).

Module 2B (Coordinate Frame): Why Population $(a_j^{\text{pop}}, b_j^{\text{pop}})$ Fails in RL

Between-Respondent Calibration vs. Within-Model Repeated-Trial Calibration



1. Between-Respondent Coordinate Mismatch
Ordering items by **external 102-LLM difficulty** b_j^{pop} yields a blurry cloud ($R^2 = 0.08$, Panel A). b_j^{pop} captures cross-architecture scale differences across 102 models, not difficulty for our single 1B policy.

2. First-Order Trajectory Jacobian Orthogonality
Linearizing the policy logit change Δz_j along update $\Delta\phi$:

$$\Delta z_j \approx J_j \Delta\phi = \alpha_{\parallel} a_j^{\text{pop}} + J_j u_{\perp}$$

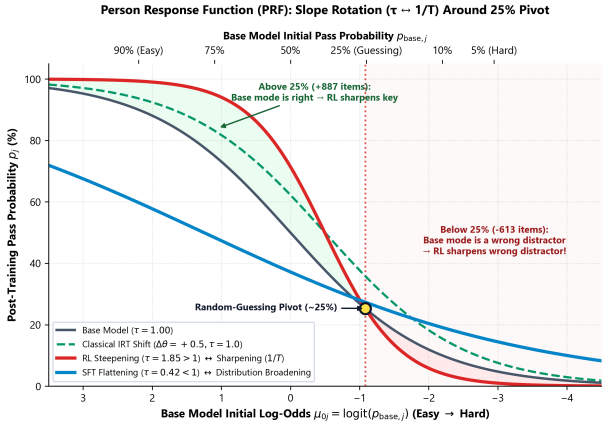
The within-model update subspace $J_j u_{\perp}$ is **orthogonal to population discrimination**:

$$\text{corr}(\mathbf{J_j u_\perp}, \mathbf{a_j^{pop}}) = \mathbf{0.072}, \quad \text{corr}(\Delta p_j^{\text{RL}}, a_j^{\text{pop}}) = +0.025$$

3. Within-Respondent Calibration $\mu_{0j} = \text{logit}(p_{\text{base},j})$
Re-ordering by the Base model's own $K = 16$ repeated-trial log-odds μ_{0j} snaps the cloud into **two opposite S-curves crossing at** $c = 1/K = 25\%$ (Panel B)!

Module 3 (Psychometric Mechanism): PRF Slope Rotation & 3PL Distractor Lock-In

Translating Temperature Scaling and RL Sharpening into Item Response Theory



1. Person Response Function (PRF) Parameterization

In psychometrics (Trabasso, 1973; Embretson & Reise, 2000), a respondent's operating curve across within-person easiness μ_{0j} is a **Person Response Function**:

$$p_{s,j} = \sigma\big(\alpha_s + \tau_s(\mu_{0j} - \mu_c)\big), \quad \mu_c = \text{logit}(0.25) = -1.10$$

- **Intercept** α_s: Horizontal knowledge shift (new support).
- **PRF Slope** $\tau_s \leftrightarrow 1/T$: Inverse decoding temperature!

2. The 3PL Distractor Trap Below $c = 25\%$

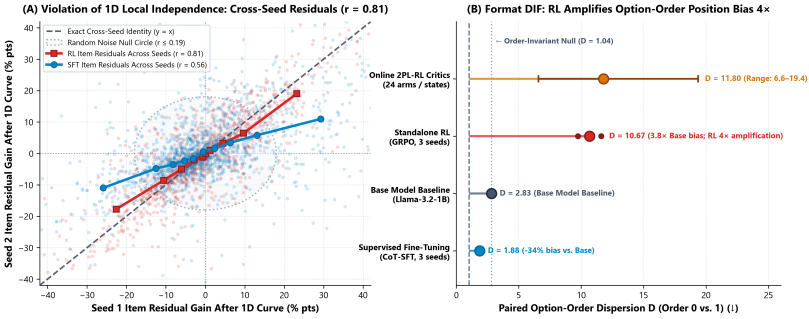
Why does GRPO ($\tau_{\text{RL}} = 1.85 > 1$) **suppress** accuracy below the Base curve on 613 items with $p_{\text{base},j} < 25\%$?

In a 4-choice item (Thissen & Steinberg, 1984), $p_{\text{base},j} < 0.25$ implies a **plausible wrong distractor** d^* holds the mode ($p_{\text{base}}(d^*) > 0.25$). Outcome RL **sharpens the wrong modal distractor** ($p_{\text{RL}}(o^*) \to 0$), destroying Pass@16!

Conversely, CoT-SFT ($\tau_{\text{SFT}} = 0.42 < 1$) lifts hard items above chance.

Module 4 (Diagnostic Added Value): Local Independence & Format DIF

Two Structural Pathologies Invisible to Aggregate Benchmark Scores

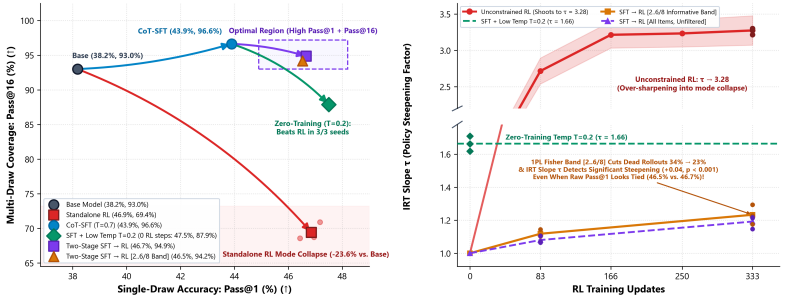


1. Violation of Conditional Local Independence ($r = 0.81$)
Under a 1D latent update, item residuals $e_j = \Delta p_j - \widehat{\Delta p}_j(\mu_{0j})$ across independent random training seeds should be uncorrelated noise ($r \leq 0.19$ null circle, Panel A). Instead, cross-seed residuals correlate at $\mathbf{r = 0.81}$ in RL ($r = 0.56$ in SFT): over 80% **of residual item variance is deterministic multi-dimensional structure** locked to the Base initialization.

2. Format DIF: $4\times$ Option-Order Position Bias ($D = 10.67$)
Swapping option order (`Order 0 vs. 1`) tests format invariance via paired dispersion D (Panel B). While `CoT-SFT` *reduces* position bias by -34% ($D = \mathbf{1.88}$ vs. $\mathbf{2.83}$ Base), `GRPO` and Online 2PL-RL critics **amplify position bias** $4\times$ ($D = \mathbf{10.67}$ and $\mathbf{11.80}$): when $\tau > 1$ sharpens low-certainty items, it locks onto letter-position heuristics!

Module 5: Testing Three Falsifiable Predictions of the Psychometric Model

Temperature Isomorphism, Two-Stage Composition, and 1PL Fisher Information Filtering



1. Temp Isomorphism ($\tau \leftrightarrow 1/T$)
Decoding `CoT-SFT` at $\mathbf{T = 0.2}$ (**0 RL steps**, $\tau = 1.66$) jumps from $43.9\% \rightarrow \mathbf{47.5}\%$ $\mathrm{Pass@1}$ while retaining $\mathbf{87.9}\%$ $\mathrm{Pass@16}$—beating standalone RL in $3/3$ seeds (Panel A)!

2. Two-Stage `CoT-SFT` → RL
First shifting support via `CoT-SFT` ($\alpha > 0$) and then applying controlled RL reaches the Pareto box ($\mathbf{46.7}\%$ $\mathrm{Pass@1}$, $\mathbf{94.9}\%$ $\mathrm{Pass@16}$) without mode collapse ($\tau = 1.19$ vs. 3.28).

3. 1PL Fisher Band $[2..6/8]$
Selecting active items with within-model Fisher info $I_j^{\text{within}} = p_j(1-p_j) > 0$ cuts dead rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$ and steepens τ_t ($+\mathbf{0.04}$, $\mathbf{p < 0.001}$, Panel B) even when $\mathrm{Pass@1}$ ties!

Synthesis: What Standard ML Metrics Miss vs. What Psychometrics Reveals

Five Structural Insights Across LLM Benchmark Evaluation and Post-Training

Stage / Arena	Superficial ML Practice (CTT)	Psychometric / IRT Lens	New Structural Finding in Our Study
1. Adaptive Eval (CAT / 1D $\hat{\theta}$)	Compress 100+ LLMs into a 1D score $\hat{\theta}$; claim 32–100 items suffice.	**Construct Validity, Factor Loading (a_s, b_s) & Domain DIF**	Population a_j^{pop} loads $5\times$ **heavier on factual recall** ($a = 1.45$) than math ($a = 0.28$), causing a $\mathbf{2.4}\%$ CAT bias floor under post-training.
2. RL Critics & Item Weighting	Plug static leaderboard $(a_j^{\text{pop}}, b_j^{\text{pop}})$ into RL advantage $\hat{A}_j$; check only Pass@1.	**Between- vs. Within-Person Frame & Equi-Accuracy (EQ)**	All 4 critics tie at $\sim 47.5\%$ Pass@1 ($\text{corr}(J_j u_{\perp}, a_j^{\text{pop}}) = 0.072$), yet diverge on **5/6 EQ item subsets** and collapse at Pass@16.
3. SFT vs. RL Post-Training	Compare aggregate Pass@1 (43.9% vs. 47.9%) and treat RL as new reasoning.	**Person Response Function (PRF) Slope τ_s & 3PL Distractors**	GRPO rotates PRF slope ($\tau = 1.85 \leftrightarrow 1/T$) around $c = 25\%$, learning **0% net new modes** ($\text{Maj@16} = 47.5\%$) and locking in wrong distractors below 25%.
4. Residual & Format Audits	Ignore item residuals and average over prompt formatting.	**Local Independence ($r_{e,e'}$) & Option-Order Format DIF (D)**	Cross-seed updates violate 1D local independence ($\mathbf{r = 0.81}$), and GRPO amplifies option-order bias $4\times$ ($D = \mathbf{10.67}$ vs. 1.88 in SFT).
5. Curriculum & Control	Heuristic rollout filtering monitored only via aggregate Pass@1.	**Within-Model 1PL Fisher Info $I_j = p_j(1-p_j)$ & (α_t, τ_t) Tracker**	Decoding CoT-SFT at $T = 0.2$ reaches $\mathbf{47.5}\%$ with 0 RL steps; Fisher band $[2..6/8]$ cuts dead rollouts $34\% \rightarrow 23\%$ and steepens τ_t ($p < 0.001$).

Core Takeaway for Psychometricians & ML Researchers: Classical IRT parameters $(a_j^{\text{pop}}, b_j^{\text{pop}})$ fail as *static training weights* not because IRT is irrelevant to LLMs, but because **cross-population differences are orthogonal to within-model learning dynamics**. Once formulated in the **within-respondent frame**, psychometric theory provides exact, falsifiable diagnostics that aggregate ML metrics miss.

Open Discussion: Where Should Our Psychometrics × LLM Agenda Go Next?

Three Concrete Questions for Group Feedback

Q1. Multi-Dimensional CAT & DIF-Aware Evaluation

Since unidimensional θ_{pop} over-selects high-a_j factual memorization items ($5\times$ skew) and breaks measurement invariance across `Base`, `CoT-SFT`, and `GRPO`:

- Should our next evaluation step be a **2D / Bi-Factor Adaptive Test** $(\theta_{\text{memory}}, \theta_{\text{reasoning}})$ with domain-stratified Fisher information constraints?
- Can we standardize **Option-Order Format DIF (D)** as a diagnostic for RL over-sharpening?

Q2. Nominal Response / 3PL Models Below $c = 25\%$

Our strongest mechanistic finding is the **polarity flip at the $c = 1/K = 25\%$ guessing threshold**, where outcome RL amplifies attractive wrong distractors:

- Can we model the full 4-option probability vector (p_A, p_B, p_C, p_D) using **Bock's Nominal Response Model (NRM)** or a **3PL distractor-repulsion penalty**?
- How should we regularize confidence sharpening when $p_{\text{base},j} \leq 1/K$?

Q3. Trajectory-Calibrated Longitudinal Item Banks

Cross-seed residuals show that 81% of item learning structure is reproducible across training runs ($r = 0.81$), just invisible to static leaderboard calibration:

- Instead of calibrating (a_j, b_j) across 102 heterogeneous leaderboard models, can we calibrate a **Longitudinal / State-Space Growth Curve Model** across checkpoints $\phi_0 \to \phi_t$?
- How few anchor items suffice to track (α_t, τ_t) online?

Thank You! Floor Open for Discussion on Psychometric Diagnostics for LLM Evaluation & Training

Backup B1: First-Order Trajectory Orthogonality & Full Experimental Table

Mathematical Details and Complete Readouts Across Post-Training Regimes

First-Order Logit Expansion Around ϕ_0

Let $z_j(\phi) = \text{logit}\, p_j(\phi)$ denote the policy log-odds on item j. Under a parameter update $\Delta\phi = \phi_T - \phi_0$:

$$\Delta z_j = z_j(\phi_T) - z_j(\phi_0) \approx \nabla_\phi z_j(\phi_0)^\top \Delta\phi = J_j \Delta\phi$$

Decomposing the update step $\Delta\phi = \alpha_\parallel u_{\text{pop}} + u_\perp$ into the 1D population-IRT direction u_{pop} and its orthogonal complement $u_\perp$:

$$\Delta p_j \approx p_{0j}(1 - p_{0j})\left[\alpha_\parallel a_j^{\text{pop}} + J_j u_\perp\right]$$

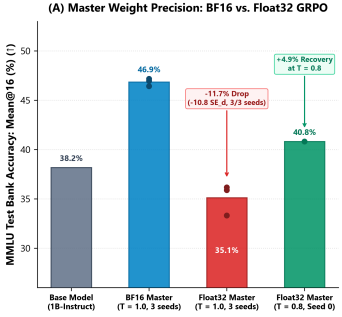
- $\text{corr}(J_j u_\perp, a_j^{\text{pop}}) = \mathbf{0.072}$ (orthogonality).
- Within-model variance $p_{0j}(1 - p_{0j})$ dominates Δp_j ($R^2 = 0.354$), whereas a_j^{pop} has near-zero alignment ($r = +0.025$).

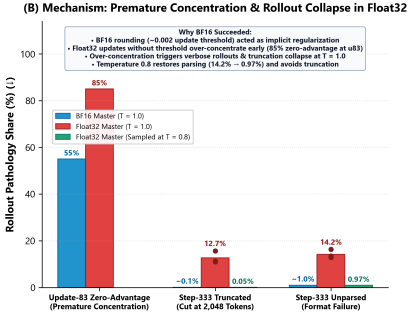
Complete Readouts on Disjoint MMLU Bank ($N = 1,472$)

Post-Training Arm	Draws	Pass@1	Pass@16	Maj@16	Order D
Base (Llama-3.2-1B)	—	38.2%	93.0%	47.6%	2.83
CoT-SFT (70B Teacher)	0	43.9%	**96.6%**	**54.6%**	**1.88**
GRPO Standard ($G = 5$)	5	46.9%	69.4%	47.5%	10.67
Base Pass-Rate Baseline	1	47.2%	58.6%	47.4%	11.42
2PL Unweighted ($\gamma = 0$)	1	46.9%	52.4%	47.1%	12.15
2PL a_j-Weighted ($\gamma = 1$)	1	**47.5%**	58.7%	47.5%	11.80
CoT-SFT @ T=0.2 (0 RL)	0	**47.5%**	87.9%	54.6%	1.95
Two-Stage SFT → RL	1	46.7%	94.9%	54.4%	3.12
Two-Stage [2..6/8 Band]	1	46.5%	94.2%	54.5%	2.98

Backup B2: Master-Weight Precision Audit (BF16 vs. Float32 GRPO)

Why BF16 Rounding Acts as Implicit Regularization Against Premature Mode Collapse





1. Does Float32 Break the $\sim$ 47% RL Ceiling? No—It Collapses Early!

Testing whether BF16 mantissa rounding ($\sim$ 0.002 update threshold) caused the Pass@1 ceiling reveals the exact opposite: removing rounding in Float32 drops accuracy by $\mathbf{-11.7}\%$ ($46.9\% \rightarrow 35.1\%$, Panel A).

2. Psychometric Confirmation of Over-Sharpening

Without BF16's implicit quantization floor, Float32 updates over-concentrate prematurely ($\mathbf{85}\%$ dead zero-advantage rollouts by update 83 vs. 55% in BF16, Panel B), triggering verbose truncation (12.7%) and format failure (14.2%).